

Dipesh Bhattarai

AI/ML Engineer

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Professional Summary

Detail-oriented AI/ML Engineer with a strong foundation in machine learning, deep learning, and data analysis. Experienced in building predictive and classification models using Python and modern ML frameworks, with hands-on exposure to neural network architectures (ANN, CNN, RNN, LSTM). Skilled in translating data into actionable insights and deploying interactive ML applications.

Technical Skills

- **Languages:** Python, C, C++, JavaScript, PHP
- **Machine Learning & Deep Learning:** Scikit-Learn, PyTorch, TensorFlow/Keras, XGBoost, LightGBM, CatBoost, ANN, CNN, RNN, LSTM
- **Data Science:** Pandas, NumPy, Matplotlib, Seaborn, Data Analysis, Data Visualization
- **Web Development:** HTML, CSS, JavaScript, React, FastAPI, Streamlit, Responsive Web Design
- **Tools & Databases:** Git, GitHub, SQL, VS Code

Projects

QuoteNext — LSTM Next-Word Predictor

Live Demo

Python, TensorFlow/Keras, Streamlit

- Trained an LSTM language model (Embedding → LSTM → Dense/Softmax) on a curated quotes dataset to predict the next word(s) in a sentence, over a ~9K-word vocabulary.
- Benchmarked against a SimpleRNN baseline to demonstrate improved long-range context retention and mitigation of vanishing gradients.
- Deployed as an interactive Streamlit app with adjustable output length and cached model loading for fast inference.

Heart Disease Prediction System

Live Demo

Python, Scikit-Learn

- Built a classification model to predict heart disease risk from medical parameters such as age, blood pressure, cholesterol, and chest pain type.
- Compared Logistic Regression, Decision Tree, and Random Forest models; improved accuracy through feature engineering.

Football Match Outcome Predictor

Live Demo

Python, XGBoost, LightGBM, CatBoost, Streamlit

- Built a web app predicting international match winners and scores using FIFA/Elo ratings and team form across 200+ national teams.
- Trained multiclass classifiers for winner prediction and regression models for goal prediction, with a live win-probability interface.

House Price Prediction API & Web App

[Live Demo](#) • [API Docs](#)

Python, Scikit-Learn, FastAPI, Streamlit

- Built an end-to-end house price prediction system using a RandomForestRegressor on the California Housing dataset, exposing predictions through a FastAPI REST API with Pydantic validation and single/batch prediction endpoints.
- Developed and deployed a Streamlit frontend and FastAPI backend as independent services, connecting them over HTTP for a live prediction application.

COVID-19 Data Analysis

[GitHub](#)

Python, Pandas, NumPy, Matplotlib, Seaborn

- Performed exploratory data analysis on global COVID-19 case, recovery, and mortality trends using Pandas and NumPy.
- Built visualizations with Matplotlib and Seaborn to uncover trends and communicate insights across regions and time periods.

E-Commerce Website

HTML, CSS, JavaScript, React

- Designed and developed a responsive e-commerce web application with product listings, shopping cart functionality, and dynamic user interactions.

Education

Tribhuvan University

2023 – 2027

Bachelor of Science in Computer Science and Information Technology (BSc CSIT)

- Relevant Coursework: Artificial Intelligence, Data Structures, Database Management Systems, Operating Systems, Computer Networks, Web Technology

Soft Skills

Analytical Thinking • Problem Solving • Team Collaboration • Adaptability • Time Management • Attention to Detail • Critical Thinking

Additional Information

- Strong interest in Artificial Intelligence, Machine Learning, and Data Science.
- Passionate about developing practical AI-driven solutions for real-world problems.
- Continuously learning modern technologies and software development best practices.